burrow in grassland soils. Without the elephants, these animals would soon disappear. Elephants are also very important seed dispersers; undigested seeds pass through their gut, are then defecated, and later germinate. Up to one-third of all West African tree species depend on elephants for their seed dispersal. Elephants also dig and maintain waterholes, which benefit many other species.

Forest-dwelling Asian elephants have a similar role. In southeast Asia, they smash through gaps and clearings in woodland, opening up holes in the canopy. The new plants that grow in these unshaded areas add to the forest's plant and animal diversity and also help a broader range of animals to thrive there.

Keystone predators

The sea otter is a marine mammal that lives in the Pacific coastal waters of North America. In the 18th and 19th centuries, they were hunted extensively for their fur. By the early 20th century, they had been wiped out in many areas, and their total population was thought to be fewer than 2,000 individuals. Since 1911, legal protection has led to a slow increase in numbers.

Sea otters are important because they eat large numbers of sea urchins. These seafloor-dwelling invertebrates graze on the lower stems of kelp that grow up from the seabed, causing it to drift away and die. If the kelp disappears, however, so do the many other marine invertebrates that graze on it. "Forests" of kelp also absorb large amounts of atmospheric carbon dioxide and, by slowing water currents, help protect coastlines from storm surges. The protection that sea otters offer kelp along stretches of open coast is therefore particularly significant.

Unlike the sea otter, some keystone species are also "apex" predators at the top of the food chain, such as the gray wolf. Before 1995, there had